1. Abstract

This report describes the design, simulation and fabrication of a spur gear reversing mechanism. The mechanism is developed using a lever operated cluster gear arrangement to reverse the direction. The project was aimed at studying the working principle of gear-based direction reversal and creating a working physical prototype that shows this mechanism. The mechanism was designed in SolidWorks and the spur gears, shafts, cluster gear, and lever components were modeled according to standard gear design calculations to ensure proper meshing and functional fit. To assess the mechanical behavior of the gear system under loading, finite element simulations were conducted individually for forward and reversing configurations to study the stress concentration, von Mises stress and deformation of the critical components. The simulation results indicated that the maximum Von Mises stress in the forward configuration was 493.2 MPa and maximum deformation was 0.0433 mm and the maximum Von Mises stress in the reversing configuration was 99.73 MPa and maximum deformation was 0.0229 mm, all of which were lower than the modeled Alloy Steel's yield strength and so the design was within safe operating limits. After designing and simulating the complete reversing mechanism, it was 3D printed on a Bambu Lab P1S with PLA+ filament and assembled. The assembled prototype was tested to verify the ability to reverse the direction of the mechanism through the shifting of the cluster gear with the lever, and the mechanism successfully performed the reversing function to confirm the design met its intended objective, except for the fitment of the printed bearings that did not rotate freely. The project, in general, shows a complete workflow from design and simulation to fabrication.

2. Introduction

Mechanical power transmission is a basic component of virtually every machine and the method by which motion and power generated at one location, such as a motor or engine shaft, can be moved to another shaft where it is actually needed to perform useful work. One of the most common and reliable elements used for this purpose are gears (Cao et al., 2022). They transmit rotational motion and torque between shafts through tooth-to-tooth contact, allowing precise control of speed, torque and direction without relying on friction or slip as do belts or chains (Ryder & Bennett, 1990). This reliability and precision are why gears form the foundation of many mechanical systems, from small household appliances to industrial machinery and automotive drivetrains (Tsvetanov et al., 2023).

Spur gears are the simplest and most widely used of all types of gears. They have straight teeth cut parallel to the axis of rotation and are mounted on parallel shafts. The principle of operation is simple (Rider, 2015). The driving gear rotates and its teeth mesh with the teeth of the driven gear. Torque is transferred directly between the two shafts (Gori et al., 2022). Spur gears are simple in terms of geometry and motion; thus they are easy to design, manufacture and analyze (Osakue, 2016). They also make a suitable and accessible starting point for the study of the basic principles of rotational motion and power transmission in an undergraduate engineering project.

One useful property of gear systems is that you can change the direction of rotation of the output shaft by changing the manner in which the gears are arranged in the mesh path (Koch, 2023). In a simple pair of spur gears in direct mesh the output turns opposite to the input, and you can restore or reverse this direction by adding or removing a gear in the path. A mechanism that uses this property to intentionally change the rotation of the output to one of the two directions without changing the rotation of the input is called a reversing mechanism (Lee, 2026). The project described here is concerned with the design and construction of one such mechanism, using a lever operated arrangement to move a cluster gear to reverse the direction of the output shaft as required.

The design and analysis of a mechanical system before it is physically built is important for the system to function correctly and safely under load. The first stage of the project was to design the reversing mechanism using Computer Aided Design (CAD) software. This enabled the gears, shafts and lever components to be accurately modeled and checked for fit and meshing. After the design is done, simulation is carried out to see the mechanical behavior of the mechanism, mainly concerning stress concentration, Von Mises stress and deformation under the load. This step was necessary to ensure that the components would be robust enough to work reliably before committing to the fabrication.

The parts of the mechanism were produced by 3D printing after the design and simulation phases. This approach was selected as it allows the translation of a CAD model directly into a physical part without the need for tooling or specialized machining setup, providing a fast, flexible and cost-effective way to produce prototype components, especially for a mechanism that uses multiple custom-shaped parts, such as gears and a shifting lever.

3. Literature Review

A spur gear is a wheel with teeth cut parallel to the axis of the wheel (Smajic et al., 2021). Spur wheels are used for transmitting rotational motion and power between parallel shafts. Its geometry is defined in standard terms such as pitch circle, module, addendum, dedendum and pressure angle (Mallesh, 2015). When the driving gear rotates, its teeth mesh directly with the driven gear, transferring torque between the shafts; since the mesh is external, two meshing spur gears always rotate in opposite directions. Due to their simplicity, low cost, high efficiency (95 to 99 percent) and absence of axial thrust on the shafts, spur gears are widely used in machines such as conveyors, automotive systems and home appliances (Rider, 2015). The main disadvantages are noise and impact loading at higher speed as the full tooth width engages suddenly which can lead to fatigue in time (Petrescu et al., 2017).

A gear train is an arrangement of two or more gears, which transmit motion from input shaft to output shaft with change of speed, torque or direction (Margielewicz et al., 2019). The gear ratio is determined by the number of teeth or pitch diameters of the meshing gears and defines the relationship between the input and output speed (Kushwaha, 2025). The direction of rotation depends on the way the gears are arranged to mesh. A simple pair of external gears will reverse the direction, but placing an idler gear between the two will restore the original direction (Kushwaha, 2025). Most reversing mechanisms work on the principle that when a gear is placed into or removed from the mesh path the direction of output is reversed. A common practical way is to mount a cluster gear on a shaft, and shift it axially with a lever, to bring different gears of the cluster into mesh with the mating shaft, producing either a direct or a reversed output (Remizovich & Abdulaeva, 2020). The working principle of this lever operated cluster gear shifting is used in this project.

When two gear teeth mesh, the force transmitted has a radial and a tangential component, the latter causing bending at the tooth root (Lee et al., 2026). The Lewis equation is a classical approach to estimate the bending stress of a tooth in which the tooth is considered as a cantilever beam loaded at the tip(Lisle et al., 2017). The simple model does not take into account the stress concentration at the tooth root filet where the actual stress is greater than the estimated. Contact (Hertzian) stress also occurs on meshing tooth surfaces (Osakue & Anetor, 2022). This is associated with surface wear and pitting rather than tooth breakage. Evaluation of these stresses is important because an under designed gear can fail by root cracking, tooth breakage or surface wear even if it appears adequate by simple calculation (Thunuguntla et al., 2024).

Finite element analysis (FEA) is a numerical method to divide a component into smaller elements to predict stress, deformation and other mechanical behavior under load, before the part is actually manufactured (Barshikar et al., 2024). The FEA can be used to obtain the real tooth geometry with root filet under realistic load conditions, which is not possible with simplified formulas, and can be used to find the critical stress regions before manufacturing (Sojčić et al., 2023). In gear analysis this is usually the Von Mises stress, an equivalent stress value used to predict yielding bending stress at the tooth root and total deformation, the displacement of the component under load. These outputs help confirm if a design is safe before it is built, and the results obtained for this project are presented in the Methodology and Findings sections below (Mohammed et al., 2025).

3D printing constructs a part layer-by-layer from a CAD model, typically with Fused Deposition Modeling (FDM) where a heated thermoplastic filament is extruded to generate each layer (Dogea et al., 2023). It's widely used for prototyping mechanical parts because it converts a design directly into a physical part without tooling or machine setup, which makes it fast, cheap and easy to iterate (Al-Ahmari et al., 2021). This makes for quick fabrication and redesign for a small mechanism like a gear and lever assembly if there are any fit issues. FDM components, however, usually have dimensional tolerances of about ±0.1 to ±0.3 mm, visible layer lines and a rougher surface finish than machined parts, and lower mechanical strength than metal components. These characteristics are relevant to the interpretation of the fabrication and assembly results discussed later in this report.

4. Methodology

4.1 Design of the Spur Gear Reverse Mechanism

The mechanism was designed as a spur gear-based mechanism that can transfer rotational movement from an input shaft to an output shaft, with the ability to reverse the direction of rotation of the output rotation by a gear arrangement operated by a lever. Instead of a separate reversing gear box, the forward and reversing paths are combined into one compact unit and the path through the transmission is selected by shifting a movable gear arrangement.

The whole mechanism consists of a series of spur gears (including the cluster gear pair, an idler gear and a single gear pair), 3 main shafts, angular contact ball bearings to support the rotating shafts, a base/frame to hold the components together in their correct relative positions and a lever operated gear changing arrangement made up of a handle, a body gear changer and a shaft adjuster. The gears are the actual means of transmitting the motion and changing the direction of the motion. The shafts support the gears and transmit the motion. The bearings support and guide the shafts, permitting them to rotate freely with a minimum of friction. The frame provides a rigid structural base for the entire assembly. The lever arrangement permits the operator to select the forward or reversing conditions. The entire assembly was modeled as a single CAD model in

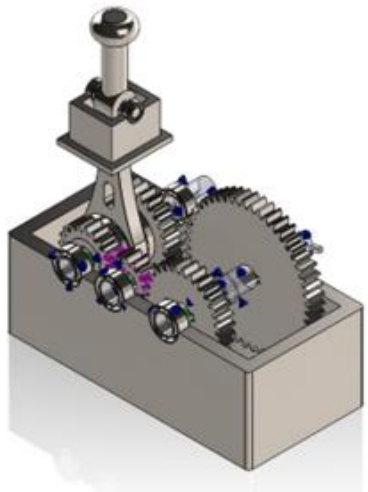


Figure 1: CAD model of the complete spur gear reversing mechanism

SolidWorks and all the components were fitted and aligned and verified for proper meshing before analysis.

4.1.1 Forward Configuration

The forward arrangement is based on the meshing of three spur gears for transmission of rotation from the input side to the output side. The input gear of the mechanism is carried on the cluster shaft. This cluster gear when in the forward position meshes with the idler gear carried on the idler shaft. The idler then meshes with the gear on the single gear shaft which in this set up forms the output side of the mechanism. Thus motion is transmitted in order from the cluster of gears at the input, through the idler gear, to the single gear on the output shaft, the idler acting as an intermediate between input and output gears without actually meshing them together.

Each of the cluster shaft, idler shaft and single gear shaft has its gears mounted on it and transmits torque along the length of the mechanism, while angular contact ball bearings along these shafts allow them to rotate freely and yet be properly supported and aligned within the frame. The base/frame holds the shafts and bearings in their fixed relative positions, so that the three gears continue to mesh properly during operation. In this configuration, the rotation at the input cluster gear causes the output shaft (single gear shaft) to rotate in the forward direction, according to the meshing sequence of the three gears described above.

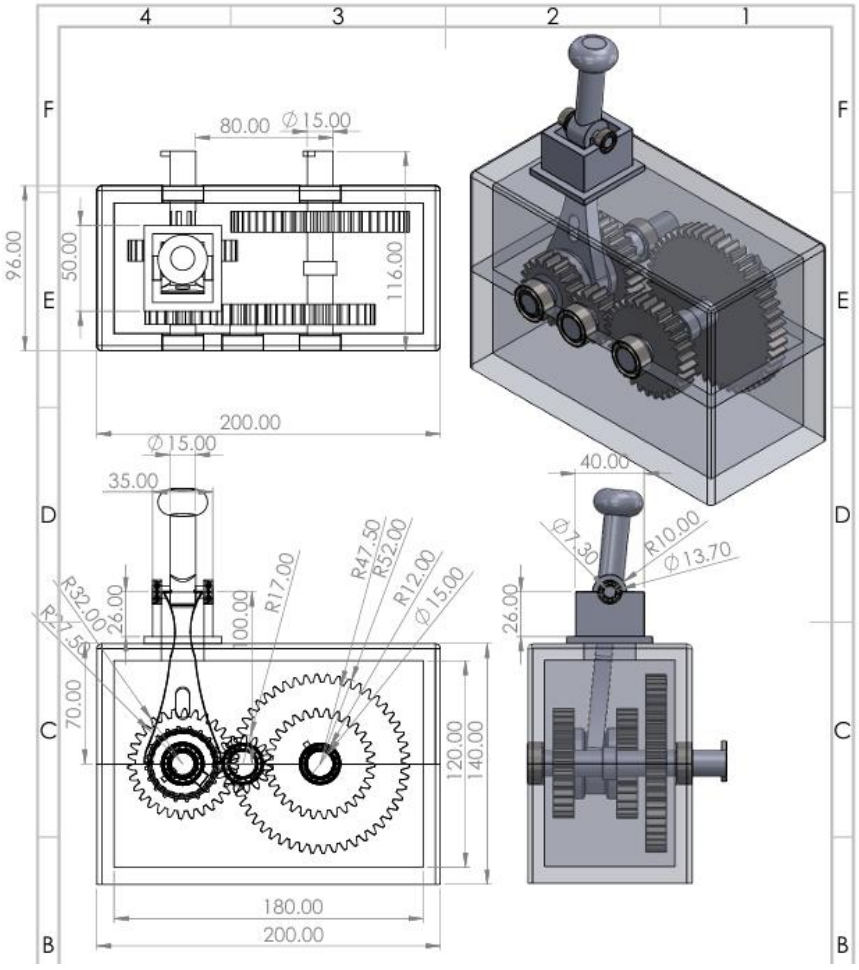


Figure 2: Forward configuration of the mechanism

4.1.2 Reversing Configuration

The reversing arrangement is located at the rear of the mechanism and involves shifting the position of the cluster gear along the cluster shaft using the lever-operated changer. Since the cluster gear also serves as the input gear, shifting its axial position changes which gear it meshes with next, thereby changing the transmission path while the direction of the input rotation itself stays the same.

In the forward position, the mechanism remains engaged through the cluster gear (input)–idler gear–single gear (output) path described in Section 4.1.1, so the output rotates in the forward direction. When the lever is moved to the reversing position, the body gear changer shifts the cluster gear along the cluster shaft, disengaging it from its forward meshing position and bringing it into a different meshing arrangement with the rear gear set. This changes the transmission path through the mechanism, and as a result, the output shaft rotates in the opposite direction compared to the forward configuration, while the input rotation applied at the cluster gear remains unchanged.

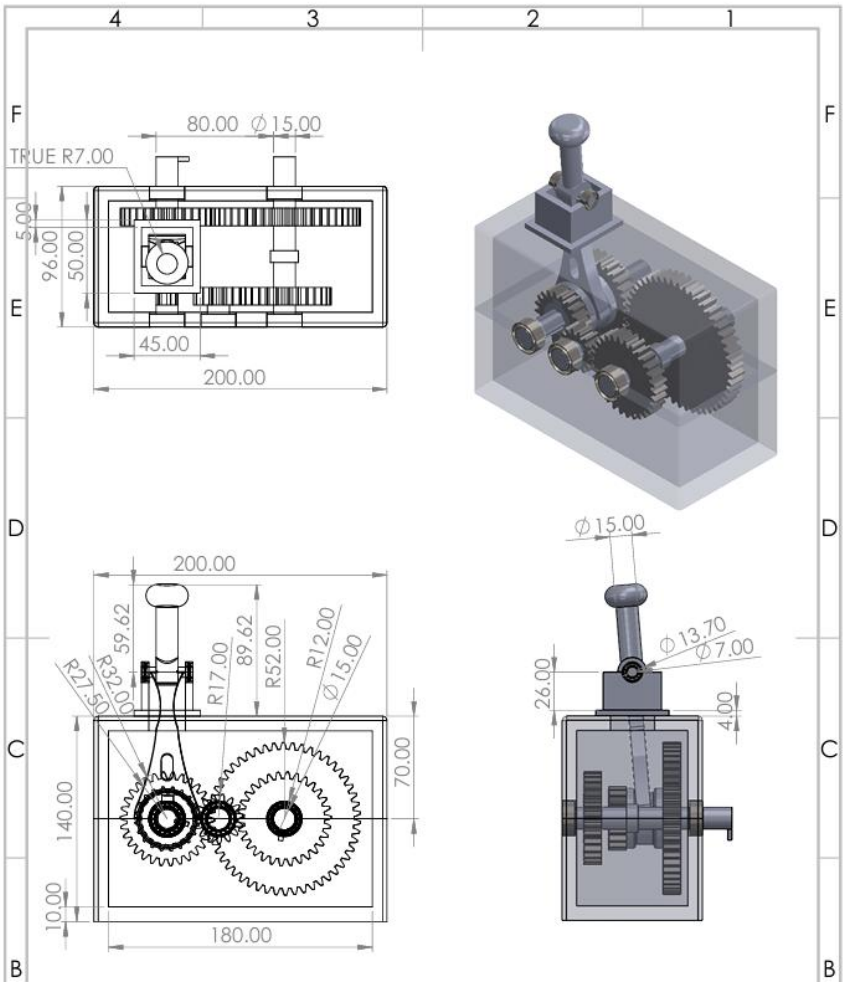


Figure 3: Reversing configuration of the mechanism

4.1.3 Lever Configuration

The lever is the main control element of the reversing system and is directly operated to select the desired operating condition. The body gear changer shifts the location of the cluster (input) gear over the cluster shaft. The lever motion is transmitted via the shaft adjuster. Therefore, the choice of the axial position determines which gears mesh and the mechanism can be shifted between the forward and reversing transmission paths as described above. Therefore, the lever acts as a simple position-selection control and not as a friction clutch as it changes the engaged gears and not the engagement or disengagement of the drive by means of friction contact.

4.2 Simulation of the Designed Mechanism

The whole mechanism was modeled in forward and reverse configurations separately and analyzed with SolidWorks Simulation, so that the mechanical response of the mechanism to each operating condition could be evaluated before its construction. In either case a static structural analysis was performed on the entire assembly, i.e. each configuration was analyzed under a constant applied load, not a time dependent load.

4.2.1 Material Assignment

Three different materials were assigned to the components of the assembly, depending on the function of each component. The gears and shafts (cluster shaft, cluster and single gears, idler gear and shaft and the shaft adjuster) were modeled as Alloy Steel, since these components are the direct carriers of the transmitted torque and are the main components of interest in the stress analysis. The base part, which forms the supporting frame, was modeled as Gray Cast Iron. This is because the base frame is a rigid structural support and not a load transmitting element. The bearing balls were modeled using Chrome Stainless Steel, the standard material for rolling-element bearings. These material assignments were consistent for forward and reverse simulation studies.

Property	**Alloy Steel (gears, shafts)**	**Gray Cast Iron (base)**	**Chrome Stainless Steel (bearings)**
Yield strength	620.42 MPa	—	172.34 MPa
Tensile strength	723.83 MPa	151.66 MPa	413.61 MPa
Compressive strength	—	572.17 MPa	—
Elastic modulus	210 GPa	66.18 GPa	200 GPa
Poisson's ratio	0.28	0.27	0.28
Density	7700 kg/m³	7200 kg/m³	7800 kg/m³

Table 1: Material properties used in the simulation

4.2.2 Boundary Conditions and Applied Load

For each study, the relevant faces of the base part were constrained with a Fixed Geometry condition, which is the part of the structure that remains fixed and provides the reference support for the rest of the assembly during operation. The load case used for the evaluation of the mechanical response of the mechanism in each configuration was a torque of 50 N·m applied at the input gear face in both configurations of forward and reverse. All parts of the assembly were considered bonded to each other in the present analysis.

4.2.3 Meshing

Both assemblies were discretized using a solid mesh generated with a blended curvature-based mesher, which automatically refines the mesh around curved and geometrically complex regions such as gear teeth, where stress tends to concentrate. A high-quality mesh setting was used for both studies.

4.2.4 Simulation Parameters Evaluated

Using this setup, the mechanical behavior of both forward and reverse configurations was evaluated in terms of the Von Mises stress distribution on each assembly, the resultant displacement (deformation) of the components under the applied torque and the equivalent strain developed in the comparison of these parameters for the two configurations gives a measure of the relative severity of loading in the forward and reverse operating conditions. The specific values obtained for each configuration are provided in the Findings section.

4.2.5 Governing Equations

SolidWorks Simulation uses the von Mises (equivalent) stress as its default failure criterion for the Alloy Steel components. In terms of the principal stresses σ_1, σ_2, and σ_3 at a point, this is defined as:

$$\sigma_v = \sqrt{\frac{1}{2}[(\sigma^1 - \sigma^2)^2 + (\sigma^2 - \sigma^3)^2 + (\sigma^3 - \sigma^1)^2]}$$

A component is predicted to yield when σ_v reaches the material's yield strength, σ_y.

Based on this criterion, the factor of safety (FOS) at the most critically stressed location is calculated as:

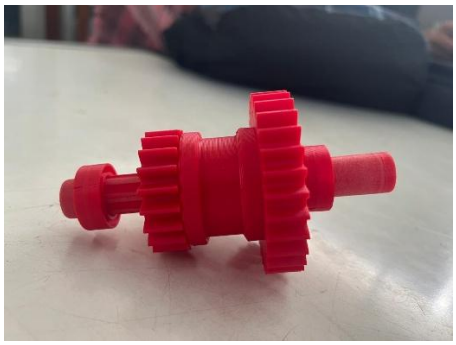
$$\text{FOS} = \frac{\sigma_y}{\sigma_{v,\ \max}}$$

where σ_v,max is the maximum von Mises stress obtained from the simulation.

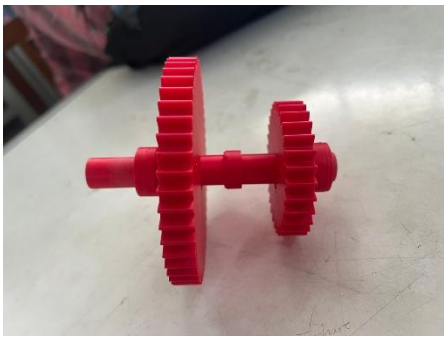
4.3 Fabrication of Components

After design and simulation all parts of the mechanism were fabricated by additive manufacturing (3D printing) instead of conventional machining, which enabled the direct production of gears, shafts and housing parts from the final CAD models. It was fabricated on a Bambu Lab P1S printer using the Fused Deposition Modeling (FDM) process.

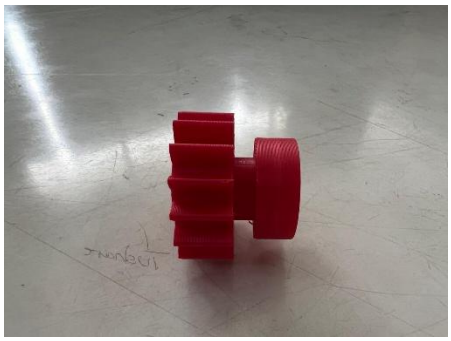
The components were printed with PLA+ filament, a biodegradable thermoplastic material, selected for its ease of printability, good dimensional stability and sufficient mechanical strength of a prototype mechanism of this size. The layer height was 0.2 mm to balance printing resolution and printing time. The parts were printed with an infill density of 15%, which provides enough internal strength while lowering the material usage and print time. A 0.4 mm nozzle diameter was used throughout which is standard for general purpose FDM printing of small to medium sized components.



1.. Input Gear (Cluster Gear)

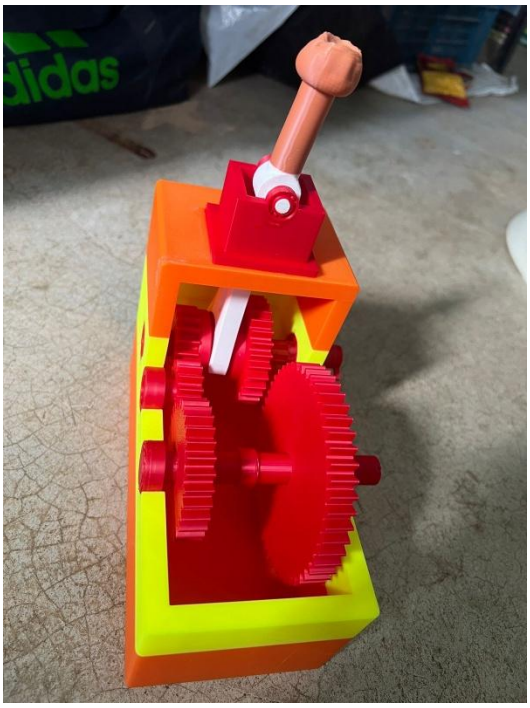


2. Output Gears

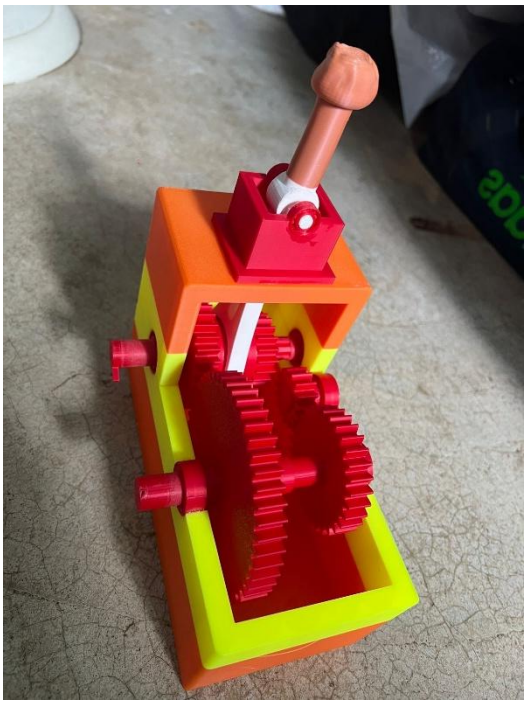


3. Idler gear

Figure 4: 3D printed Parts of the Mechanism



1.Forward Mechanism



2. Reversing Mechanism

Figure 5: 3D Printed Assembled Mechanism

All components were oriented flat on the print bed to improve bed adhesion and dimensional accuracy, and tree-type support structures were used where necessary to support overhanging features without excessive material use or difficulty during support removal. It took about 13 hours of total print time to make the full suite of parts needed.

Parameter	Value
Printer	Bambu Lab P1S
Process	Fused Deposition Modelling (FDM)
Material	PLA+ (biodegradable)
Layer height	0.2 mm
Infill density	15%
Nozzle diameter	0.4 mm
Print orientation	Lay flat on bed
Support structure	Tree supports
Total print time	~13 hours

Table 2: 3D printing parameters

5. Findings

5.1 Simulation Findings

The static simulation results for the forward and reverse configurations are summarized in Table X and provide a basis for comparison of the mechanical severity of the two operating conditions.

Parameter	Forward configuration	Reverse configuration
Maximum Von Mises stress	493.2 MPa	99.73 MPa
Maximum resultant deformation	0.0433 mm	0.0229 mm
Maximum equivalent strain	1.884×10^{-3}	4.126×10^{-4}
Yield strength (Alloy Steel)	620.42 MPa	620.42 MPa

Table 3: Summary of Simulation Results

Maximum Stress: The maximum Von Mises stress in the forward configuration was determined to be 493.2 MPa at a node on one of the meshing gears which transmits the torque. This value is less than the yield strength of the Alloy Steel material (620.42 MPa) giving an approximate factor of safety of 1.26, which shows that the forward configuration is working reasonably close to its safe stress limit under the applied 50 N.m torque. In reverse configuration, maximum Von Mises stress was significantly lower (99.73 MPa), resulting in a much higher factor of safety of approximately 6.22. This difference indicates that the reverse configuration has a lower effective loading on the gear teeth for the same input torque, most likely due to the different gear pair that takes the load in this configuration.

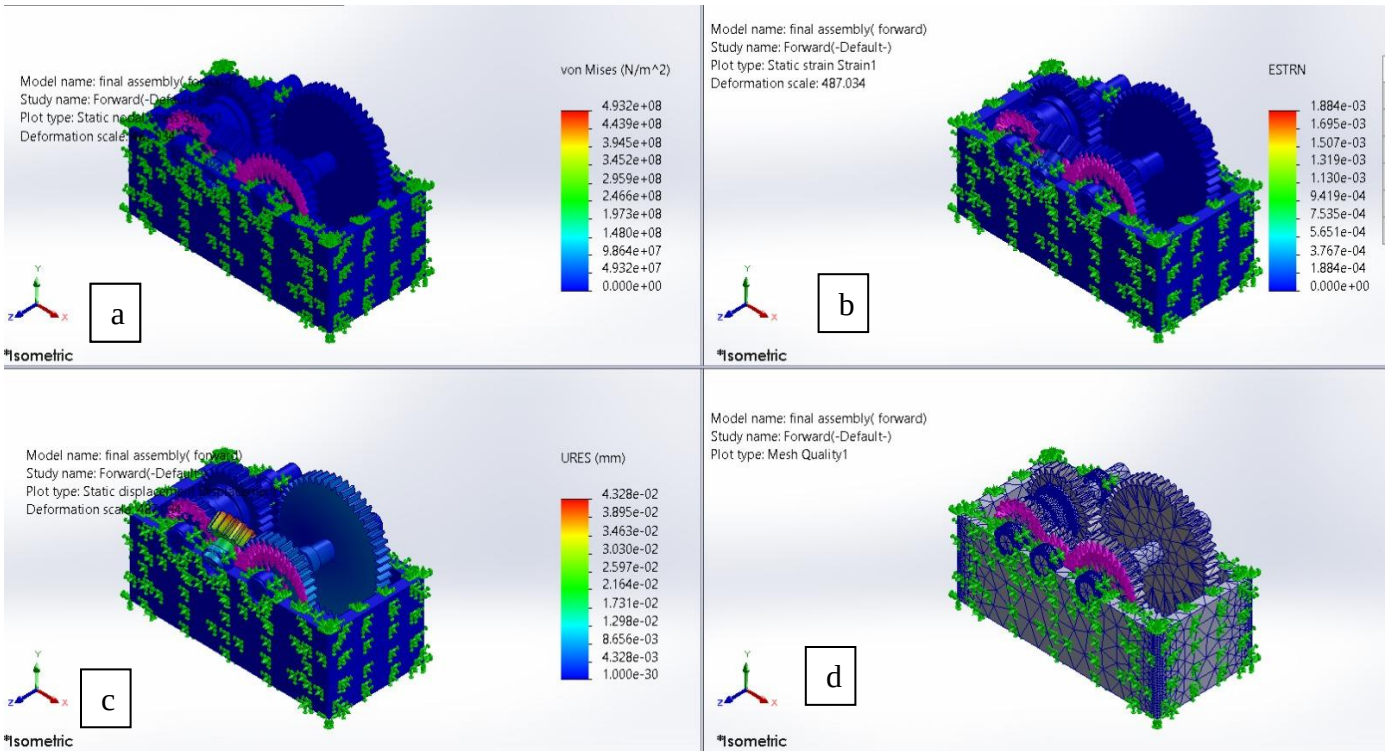


Figure 6: Simulation results and mesh of the forward configuration: (a) von Mises stress, (b) equivalent strain, (c) resultant displacement, and (d) mesh quality.

Maximum deformation: The maximum resultant displacement was 0.0433 mm for the forward configuration and 0.0229 mm for the reverse configuration. In both cases the deformation was in the direction away from the fixed support region, which is consistent with the expected behavior of a shaft/gear system loaded in torsion. In both instances the values are small when compared to the overall size of the components, which suggests that the mechanism remains dimensionally stable under the applied load in either configuration.

Maximum equivalent strain: The maximum equivalent strain recorded was 1.884×10^{-3} in the forward configuration and 4.126×10^{-4} in the reverse configuration.

Additional observation: In the forward configuration's deformation plot, the idler gear appears visibly shifted away from its meshed position relative to the other two gears.

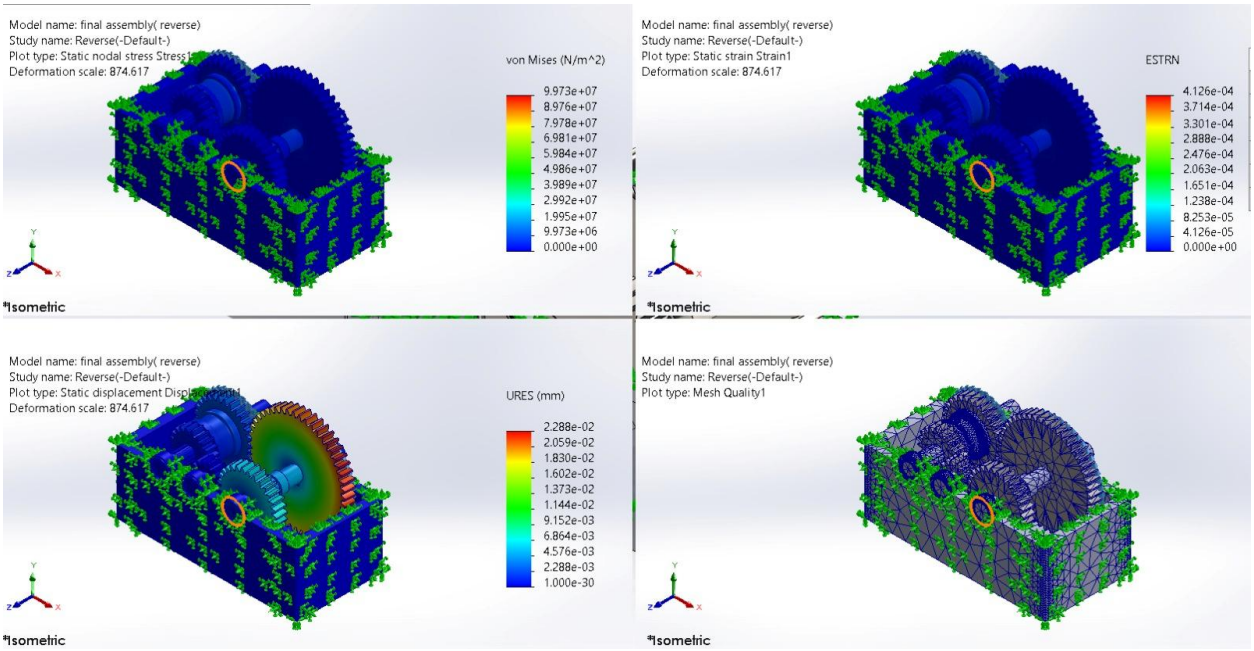


Figure 7: Simulation results and mesh of the Reversing configuration: (a) von Mises stress, (b) equivalent strain, (c) resultant displacement, and (d) mesh quality.

5.2 Fabrication and Assembly Findings

Most of the printed parts fit together well during assembly, being close enough to their CAD dimensions, with gears, shafts, and the base parts needing only minor cleanup by hand before assembly. The shafts were properly aligned in their own bores in the base part and gear-changer parts. The frame kept the shafts in their proper relative positions without any apparent misalignment.

Assembly was followed by a check of gear engagement. In the forward lever position the cluster gear, idler gear and single gear engaged correctly and transmitted rotation in the forward direction. In the reverse lever position the cluster gear was correctly shifted into the rear gear arrangement and reversed the output direction as it was supposed to.

One fitting problem occurred with the printed bearing components: the support material generated during printing became lodged inside the bearing races. Attempts were made to remove this support material after printing, but the bearings did not rotate freely even after cleanup, indicating that the internal geometry of the printed bearings was not fully functional as a rolling-element support.

6. Discussion

The simulation results show a significant difference in the mechanical severity of the forward and reverse configurations. For the forward configuration, the maximum Von Mises stress is 493.2 MPa while for the reverse configuration it is only 99.73 MPa with the same input torque of 50 N·m. Most likely this difference is due to which gear pair is carrying the load in each configuration. In the forward path the torque passes through the cluster gear, the idler gear and the single gear in sequence, whereas in the reverse path the torque is transmitted through a different gear pair at the rear of the assembly. If this rear gear pair has a larger tooth size, larger pitch diameter or more favorable contact geometry than the forward path gears, then the same input torque would result

in a lower tangential tooth load and hence lower stress. This also explains the difference of factor of safety which is much higher for reverse configuration (≈6.22) than for forward configuration (≈1.26), showing that the forward path is the more critical loading condition for the overall design and needs to be prioritized if any redesign or reinforcement is to be considered.

The trend of the maximum equivalent strain is similar to the stress results, but it has a greater value (1.884 x 10-3) for the forward configuration than the reverse configuration (4.126 x 10-4), which is consistent with the higher stress carried by the forward configuration. The material was modeled as linear elastic, so strain in this analysis scales directly with stress, rather than indicating any plastic deformation; both strain values remain well below the strain level at which Alloy Steel would begin to yield, which is consistent with the calculated factors of safety and confirms that neither configuration is predicted to undergo permanent deformation under the applied 50 N·m load.

The apparent movement of the idler gear in the deformation plot for the forward configuration is likely a visualization artifact, rather than any actual physical displacement of the gear away from its meshed position. We show displacement plots from SolidWorks Simulation (with an exaggerated deformation scale) to reveal displacements that are very small otherwise; the forward study used a deformation scale of 487×, which is nearly 500 times the actual displacement (roughly 0.04 mm) for the plot. At this scale, a slight relative displacement between meshing components can be seen as a visible gap or shift in the rendered image, although the actual physical displacement is minor and gear meshing was not lost. This should be verified by looking at the deformation plot on a scale close to 1:1, or by checking the actual displacement values at the mesh point of the idler gear, and not just on the scaled plot that is visually.

The largest single gear has a larger forward direction deformation, which is consistent with its position in the assembly relative to the fixed support. The gear is the furthest from the fixed base along the transmission path and so has the largest accumulated torsional and bending deflection of any of the shafts and gears upstream of it. Therefore the displacement across the face of the gear is visibly larger and varies more smoothly than for components further up the transmission path toward the fixed support. This is normal behavior for a shaft and gear system under torsional loading and does not in itself indicate a design deficiency. The absolute displacement values involved (well under 0.05mm) are very small in relation to the size of the components.

The maximum stress locations for both configurations are at the meshing gear teeth which is in agreement with the general principle discussed in Literature Review, that stress is concentrated at the tooth root under tangential loading. This is further corroborated by the secondary region of elevated stress found near a support/fillet area in the reverse configuration, as filets and geometric transitions are well known locations for stress concentration. Overall these results would suggest that the gear teeth are the most likely component to control the mechanical performance and ultimate failure of the mechanism rather than the shafts, base or frame.

7. Limitations

While the project achieved its overall goals of designing, simulating and fabricating a spur gear reversing mechanism, there are several limitations that should be noted.

Simulation Limitation: The simulation was performed as a static structural analysis with one value of applied torque (50 N.m) which is one of the operating conditions and not the full range of loads that the mechanism may be subjected to during practical use. While these are relevant to the gear performance over time, dynamic effects such as impact loading during the engagement

of gears, vibration, and repeated cyclic loading (which determines fatigue life), were not considered. In the simulation setup friction was disabled, which simplifies the analysis, but does not capture additional losses and localized heating at real gear tooth and bearing contacts. In addition, the torque of 50 N·m used in the simulation was a selected loading condition for the analysis purpose and it was not obtained from an experimentally measured operating torque of the physical prototype.

Material and modeling limitations: In the simulation, the gears and shafts were assumed to be made of Alloy Steel, while the actual fabricated parts were 3D printed in PLA+ which has significantly lower stiffness and strength than steel. Therefore, the simulated stress, deformation and factor-of-safety values are not a direct representation of the mechanical behavior of the physical printed prototype, and should be considered as an indicative structural study of the design geometry, rather than a prediction of the real performance of the printed part. The exact identity of some fixed geometry and load-application faces used in the simulation could not be fully confirmed from the report alone.This limits the precision with which boundary conditions can be described.

Fabrication limitations: During assembly, the printed ball bearing parts were not free to rotate since the printing process produced support material inside the enclosed bearing geometry that could not be completely removed. This is a known limitation of FDM 3D printing for enclosed, self-supporting mechanical features like rolling-element bearings and did have some impact on the smoothness of shaft rotation in the final assembly. The printed parts are inherently weaker than their metal equivalents meaning the mechanism's real load carrying capacity is lower than what the Alloy Steel simulation suggests.

Testing limitations: Testing of the assembled prototype was limited to verifying that the mechanism could switch between forward and reverse output rotation as intended. No quantitative measurements were taken of parameters such as actual operating torque, rotational speed, backlash, or long-term wear, so the testing confirms functional operation of the mechanism but does not provide a detailed performance characterization.

8. Conclusion

This project involved the design, simulation, fabrication and testing of a spur gear reversing mechanism that reverses the output rotation through a lever operated cluster gear arrangement. It has a three gear meshing path for the forward configuration, and a separate rear gear arrangement for reverse.

Static simulations at a torque of 50 N·m gave a maximum Von Mises stress of 493.2 MPa (forward) and 99.73 MPa (reverse), both lower than the yield strength of the modeled Alloy Steel, with the highest stresses being located at the meshing gear teeth as anticipated. All parts were 3D printed on a Bambu Lab P1S in PLA+ and assembled into a working prototype that correctly reversed rotation when the lever was used. The main problem was the printed bearings, which did not rotate freely because of trapped support material

This project successfully demonstrates a complete design to testing workflow for a spur gear reversing mechanism and confirms 3D printing as a viable fabrication method for the gears, shafts and frame of the mechanism. The bearings were found to be an area of further design work.

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